

System management

`uname -a` Displays all system information
`hostnamectl` Shows current hostname and related details
`sudo hostnamectl set-hostname <name>` Renames the computer (not the user. [user@pc_name:](#))
`df -h` Shows disk usage in a human-readable format.
`sudo systemctl start` Starts a service.
`sudo systemctl stop` Stops a service.
`sudo systemctl restart` Restarts a service.
`sudo systemctl status` Checks the status of a service.
`sudo systemctl reload` Reloads a service's configuration.
`ls` Lists files and directories.
`touch` Creates an empty file or updates the last accessed date.
`cp` Copies files from source to destination.
`mv` Moves files or renames them.
`rm` Deletes a file.
`pwd` Displays the current directory path.
`cd` Changes the current directory.
`mkdir` Creates a new directory. `-p` : create parent directories if needed; no error if exists.
`find [directory] -name` Finds files and directories.
`grep` Searches for a pattern in files.
`sudo apt install` Installs a package.
`sudo apt install -f --reinstall` Reinstalls a broken package.
`apt search` Searches for APT packages.
`apt-cache policy` Lists available package versions.
`sudo apt update` Updates package lists.
`sudo apt upgrade` Upgrades all upgradable packages.
`sudo apt remove` Removes a package.
`sudo apt purge` Removes a package and all its configuration files
`ip addr show` Displays network interfaces and IP addresses.
`ip -s link` Shows network statistics.
`ss -l` Shows listening sockets.
`ping` Pings a host and outputs results.
`ssh <user@host>` Connects to a remote host via SSH.
`scp <source> <user@host>:<destination>` Securely copies files between hosts.

User management

`w` Shows which users are logged in.
`sudo adduser` Creates a new user.
`sudo deluser` Deletes a user.
`sudo passwd` Sets or changes the password for a user.
`su <username>` Switches user.
`sudo passwd -l` Locks a user account.
`sudo passwd -u` Unlocks a user password.
`sudo chage` Sets user password expiration date.
`id [username]` Displays user and group Ids.
`groups [username]` Shows the groups a user belongs to.
`sudo addgroup` Creates a new group.
`sudo delgroup` Deletes a group
`sudo usermod -aG <group> <username>` Adds user to a group (append (do not remove existing groups)).
`sudo chgrp <group> <folder path>` Changes group ownership. Members of the group can access the directory.

Permission management

`sudo chmod [who (u;g;o)][+/-][permissions (r;w;x)]` Changes file permissions.

`sudo chmod u+x` Makes a file executable by its owner.

`sudo chmod -R .....` Recursive permission (grants permission to subfolders).

`sudo chmod 770 <folder path>` Sets directory permissions. Only owner and group can access.

770

```
|
|├─ others = 0
|├─ group = 7
└─ owner = 7
```

7 = rwx (Can read, write and run files)

4 = read (r)

2 = write (w)

1 = execute (x)

5 = r-x (Can read and run but can't write)

`chmod 770 = chmod rwxrwx---`

`chmod 750 = chmod rwxr-x---`

`chown [user]:[group]` Changes file owner and group.

ACL

`sudo apt install acl` Standard Linux permissions are limited; ACL allows per-user exceptions.

`sudo setfacl -m [who (u;g;o)]:[username/groupname]:[permission(rwx)] <folder_path/file_path>`

Overrides normal permission behavior with fine-grained control.

`sudo setfacl -m u:Jancsi:r-- /srv/postaskisasszony/mai_teendok.txt` Gives Jancsi read-only access to the file.

`getfacl <folder_path/file_path>` Displays ACL permissions.

Firewall

`sudo apt install ufw` Installs firewall management tool.

`sudo ufw status` Displays the status of the firewall.

`sudo ufw enable` Enables the firewall.

`sudo ufw disable` Disables the firewall.

`sudo ufw allow <portnumber>/<protocol (tcp/udp)>` Allows traffic on a specific port or service.

`sudo ufw deny <portnumber>/<protocol (tcp/udp)>` Denies traffic on a specific port or service.

`sudo ufw delete allow/deny <portnumber>/<protocol (tcp/udp)>` Deletes an existing rule.

HTML and HTTP

`sudo apt install apache2` Installs Apache web server.

`wget <website URL>` Downloads HTML page (if not specified the location, then it downloads it to the active path).

`curl <website URL>` Displays HTML page's source code. Similar to wget, but it doesn't create a file.

`sudo apt install squid` Installs proxy server software.

`sudo nano /etc/squid/squid.conf` Opens Squid configuration file.

SSH VirtualBox-on keresztül:

1. SSH client/server telepítése (jó eséllyel már telepítve van. Írd be, hogy ssh és üss egy enter-t.)
2. Gép>Konfigurálás>Hálózat>Kártya 1>Port továbbítás>Új port hozzáadása
3. Rule 1 [Protokoll: TCP; Gazda IP: <hagyd üresen>; Gazda port: 22; Vendég IP: <hagyd üresen>; Vendég Port: 22.]
4. VirtualBox-on kívüli gazdagépen indítsd el a cmd/powershell és írd be:
`ssh <username>@127.0.0.1`